

ISMS V3 Octopus Cable Connection Key

ISMS Housing Side			Exterior Connection Side					
Pin No.	Wire Base Color	Wire Stripe Color	Routed To	SubConn Connector	Pin Role	Pin No.	Wire Base Color	Wire Stripe Color
1	Black		Vehicle / Topside	MCIL8M	Main Power V+ 1 (VIN1)	1	Black	
2	White		Vehicle / Topside	MCIL8M	Main Power V+ 2 (VIN2)	2	White	
3	Red		Vehicle / Topside	MCIL8M	Main Power GND (VIN-)	3	Red	
4	Green		Vehicle / Topside	MCIL8M	tx/rx Arduino (tx)	4	Green	
5	Orange		Vehicle / Topside	MCIL8M	tx/rx Arduino (rx)	5	Orange	
6	Blue		Vehicle / Topside	MCIL8M	tx/rx RGA (tx)	6	Blue	
7	White	Black	Vehicle / Topside	MCIL8M	tx/rx RGA (rx)	7	White	Black
8	Red	Black	Vehicle / Topside	MCIL8M	Communication GND (shared)	8	Red	Black
9	Green	Black	Fluid Pump	MCIL5F	Fluid Pump ■ GND (Shared)	1	Black	
10	Orange	Black	Fluid Pump	MCIL5F	Fluid Pump ■ Analog (0-5 V)	2	White	
11	Blue	Black	Fluid Pump	MCIL5F	Fluid Pump ■ Power (+12v)	3	Red	
N/a	N/a	N/a	Fluid Pump	MCIL5F	UNUSED Fluid Pump ■ Tachometer	4	Green	
12	Black	White	Fluid Pump	MCIL5F	Fluid Pump ■ Fwd/Rev 0/5v	5	Orange	
13	Red	White	PH Probe	MCIL6F	LX/TX: pH Power V- (GND)	1	Black	
14	Green	White	PH Probe	MCIL6F	LX: USB Data (-) TX: RS485 Data	2	White	
15	Blue	White	PH Probe	MCIL6F	LX/TX: pH Power V+ (+5 VDC)	3	Red	
16	Black	Red	PH Probe	MCIL6F	LX:USB Data (+) TX: pH Analog Signal (0-3.3v)	4	Green	
N/a	N/a	N/a	PH Probe	MCIL6F	UNUSED LX: Shield TX: RS485 Data	5	Orange	
N/a	N/a	N/a	PH Probe	MCIL6F	UNUSED pin for Compatibility	6	Blue	

Compatibility

The 8 pin vehicle/topside connector pinout is backwards compatible with vehicles that connected to prior ISMS versions. The 16 pin main bottle connector, fluid pump, and PH probe connectors ARE NOT backwards compatible.

Locking Sleeves

All Subconn Male (MCILxM) connectors use a female locking sleeve. Conversely the Female Subconns (MCILxF) use a male locking sleeve. This gives the exposed pins a bit of protection.

pH Probe

This cable is default compatible with pyroscience aquapHOx-LX model pH logger but can be adapted to fit aquapHOx-TX or pico-ph-sub model pH sensors with an adapter cable and firmware changes. The pin role for the pH probe is specified based on how the pin would map to the connector on an aquapHOx-LX with "LX:" or aquapHOx-TX with "TX:"

Topside / In Vehicle Connector Pinout

SubConn Connector	Pin Role	Subconn Pin No.	Wire Base Color	Wire Stripe Color	Topside Interface Connector
MCIL8F	Main Power V+ 1 (VIN1)	1	Black		Bare Lead or 1x2 pin female connector (Microfit 3.0)
MCIL8F	Main Power V+ 2 (VIN2)	2	White		Bare Lead or 1x2 pin female connector (Microfit 3.0)
MCIL8F	Main Power GND (VIN-)	3	Red		Bare Lead or 1x2 pin female connector (Microfit 3.0)
MCIL8F	Main Comms RX (From TX pin at ISMS)	4	Green		To Female DB9 Serial > RS232 to USB converter > Windows 11 Computer
MCIL8F	Main Comms TX (From RX pin at ISMS)	5	Orange		To Female DB9 Serial > RS232 to USB converter > Windows 11 Computer
MCIL8F	RGA Comms RX (From TX pin at RGA)	6	Blue		To Female DB9 with 'fake-handshake' null modem > RS232 to USB converter > Windows 11 Computer
MCIL8F	RGA Comms TX (From RX pin at RGA)	7	White	Black	To Female DB9 with 'fake-handshake' null modem > RS232 to USB converter > Windows 11 Computer
MCIL8F	Secondary GND (VIN2) Previously known as 'Communication GND'	8	Red	Black	Bare Lead or 1x2 pin female connector (Microfit 3.0)

Octopus Cable Subconn Connectors Detail

Octopus Cable Subconn	Sleeve	Mated Device	Mating Subconn	Mating Sleeve	Mating Subconn Matched Dummy
MCIL16F	DLSA-M	ISMS Main	MCBH16M	DLSA-F	MCDC16F
MCIL8M	MCDLS-F	AUV/SUB Connection	MCIL8F	MCDLS-M	MCDC8F
MCIL5F	MCDLS-M	External Fluid Pump	SVS-BHC5M-M10	MCDLS-F	MCDC5F
MCIL6F	MCDLS-M	External PH Probe	MCBH6M (Part of PH Probe)	IN-BUILT	MCDC6F
Benchtop Subconn	----- TOPSIDE / BENCHTOP -----				
MCIL8F	MCDLS-F	Topside Laptop	N/A	MCDLS-M	N/A